Barthélémy Doba

Curriculum Vitæ

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Education

2024 – 2027 **PhD in Pure Mathematics**,
Sorbonne Université *(Institut de Mathématiques de Jussieu - Paris Rive Gauche (IMJ-PRG)*, Paris, France
Thesis title : *Geometric properties at infinity of minimal surfaces arising from cyclic Higgs bundles in the Hitchin component*
Supervisor : Prof. Andrés Sambarino (France)

2023 – 2024 **Master II in pure mathematics**,
Université Sorbonne Paris Nord, Paris, France
Thesis title : *Rigidité infinitésimale des variétés hyperboliques compactes à bord totalement géodésique en dimension supérieure*
Supervisor : Prof. Andrés Sambarino (France)

2022 – 2023 **Structured Master in pure mathematics**,
African Institute for Mathematical Sciences (AIMS), Limbé, Cameroon
Thesis title : *Elliptic Curves - Quantum Computing*
Supervisor : Prof. Dr. Hans Georg Rück (Germany)

2021 – 2022 **Master I in pure mathematics (Analysis, Algebra and Geometry)**,
University of Douala, Douala, Cameroon

2020 – 2021 **Bachelor degree in pure mathematics (Analysis, Algebra and Geometry)**,
University of Douala, Douala, Cameroon

Publications & Preprints

2026 **Mumford compactness for** $\mathrm{PSL}(3,\mathbb{R})$**-Hitchin representations**

2026 **Geometric properties of** $\mathrm{Sp}(4,\mathbb{R})$**-Hitchin representations**

2026 **Geometry at infinity of cyclic minimal surfaces in the Hitchin component** (in preparation)

Research interest

I am interested in minimal surfaces associated to cyclic Higgs bundles in the Hitchin component, Anosov representations, (higher)-Teichmüller theory, Thermodynamic, Affine building, Weak systems of Hodge bundles, Variations of Hodge structures and (G,P)-opers. I also have interests in Geometric analysis, Hyperbolic geometry, Chern-Weil and Chern-Simons theory

Talks & Seminars

Nov 2026 **Geometry at infinity of cyclic minimal surfaces in the Hitchin component**, *Geometry seminar*, MAX-PLANCK-INSTITUT, Leipzig, Germany

Oct 2026 **Mumford compactness for $\mathrm{PSL}(3,\mathbb{R})$-Hitchin representations**,
Research seminar of dynamics, Centro de Matemática, Universidad de la República, Uruguay

Aug 2026 **Sharpness of proper and cocompact actions on reductive homogeneous spaces**,
Workshop on Compact Clifford-Klein Forms : constructions and obstructions, Hohenwart, Germany

Oct 2026 **Cyclic Higgs bundles and critical exponent**,
Mini talk at the Workshop on Compact Clifford-Klein Forms : constructions and obstructions, Hohenwart, Germany

Jun 2026 **Cyclic Higgs bundles and critical exponent**,
Journée d'équipe ACG, IMJ-PRG, Paris

May 2026 **Le cône limite de Benoist a un interieur non vide et pourquoi cela est faux pour les corps non-archimedien**,
Groupe de travail organisé par Nicolas Tholozan, ENS, Paris

Feb 2026 **Introduction to Chern-Simons and Chern-Weil theory : applications to cohomological invariants for representation varieties**,
Groupe de travail sur un papier de Nicolas Tholozan, IMJ-PRG, Paris

Nov 2025 **Anosov representations and applications**,
Groupe de travail dynamique sauvage, Jussieu, Paris

Nov 2024 **Surfaces minimales dans les espaces localement symétriques de type non-compact**,
Rencontre Masters-Doctorants, Université Paris Cité, Paris

May 2024 **Gromov hyperbolicity of the fundamental group of a compact surface with genus at least two**,
Exposé encadré, Sorbonne Université, Paris.

Research stays

Sept-Oct 2026 **Research visit supervised by Andrés Sambarino at Centro de Matemática, Universidad de la República,** Uruguay

Conferences & Workshops attented

Oct 2026 **Worksop on Group dynamics**,
Montevideo, Uruguay, 2026/10/05 to 2026/10/09

Aug 2026 **Worksop on Compact Clifford-Klein Forms : constructions and obstructions**,
Hohenwart, Germany, 2026/08/24 to 2026/08/28

May 2026 **Workshop on Character varieties under the lens of Hodge theory**,
Leipzig, Germany, 2026/05/04 to 2026/05/08

Jan 2026 **Winter School Côte d'Azur : Geometric Analysis and Surface Groups**,
Université Côte d'Azur, Nice, France, 2026/01/05 to 2026/01/09

Sept 2025 **Workshop on Groups, dynamics and surfaces**,
Invited by Bram Petri, Jussieu, France, 2025/09/17 to 2025/09/19

Aug 2025 **Workshop on Universal Teichmüller Theory : Classical and Higher-Rank**, Aussois, France, 2025/08/25 to 2025/08/29

April 2025 **Workshop on Geometric structures and discrete group actions**, CIRM (Marseilles), France, 2025/04/14 to 2025/04/18

April-Jul 2025 **Summer school followed by conferences on Higher rank geometric structures, Higgs bundles, physics and Low-dimensional phenomena: geometry and dynamics**, IHP, France, 2025/04 to 2025/07

March 2025 **ARN project on Autour de la Géométrie Hilbert sur les corps valués**, Invited by Antonin Guilloux, Jussieu, France, 2025/03/24 to 2025/03/27

Aug 2026 **Rencontre des jeunes chercheurs africains en France**, IHP, France, 2024/12/12 to 2024/12/13

Oct 2024 **Conference on 20 years of Anosov representations**, Leipzig, Germany, 22024/10/7 to 2024/10/11

Rewards

Jun 2024 **Doctoral contract at Sorbonne Université**

Jun 2025 **200 euros at Université Sorbonne Paris Nord (USPN) as the valedictorian of my class (academic year 2023-2024) for the Master II program in Fundamental Mathematics**

2024 **Selected by the Italian program led by Marco Gurati for a mathematical program that ran from March 1st to June 30th**

2023-2024 **Scholar of the Paris Graduate School for Mathematical Sciences (PGSM) at Université Sorbonne Paris Nord (USPN)**

Jun 2023 **300.000 FcFa at Aims-Cameroon as the valedictorian of my class (academic year 2022-2023) for the Structured Master in Fundamental Mathematics**

Feb 2023 **A book received as the vice best student at the mathematical games organized at AIMS-Cameroon by the Emeritus Professor Michel Waldschmidt from Sorbonne University**

2022-2023 **Scholar of the Structured Master program at Aims-cameroon in Fundamental Mathematics**

2021-2022 **100.000 Fcfa at the University of Douala given by the President of the Republic of Cameroon as the top of my class in my Bachelor's (2020-2021) and Master's I (2021-2022) programs in pure mathematics**

Teaching

Sept-Dec 2026 Analyse II, level 2, Sorbonne Université, France

Jan-May 2026 Analyse II, level 2, Sorbonne Université, France

Jan-May 2026 Analyse fonctionnelle, Bachelor level, Sorbonne Université, France

2020-2023
1. Differential Calculus, Algebras and Banach Spaces, level 3
2. Partial Differential Equations, level 3
3. Functional Analysis and Topological Vector Spaces, Master I level
4. Distribution Theory, Master I level

5. Introduction to Riemannian Geometry, Master I level
6. General Topology, Bachelor level
7. Measure Theory and Integration, Bachelor level
8. Projective Geometry, Bachelor level
9. Numerical Analysis, Bachelor level

Programming

- Sage Math
- Python
- Octave
- R-Studio

Languages

- French (Mother tongue)
- English (Fluent)
- Spanish (Basic)